



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

AIML for Robotics

Module 4 – Mapping (Part 3)

SLAM

Session Faculty:

Raja vadhana Prabhakar

The course owner and the authors of this course deck, Prof. Raja vadhana and Dr.Sangeetha, are gratefully acknowledging all authors who made their course materials freely available online.

Detailed demo on the role of ML/ DL/RL techniques in Localization was the first half of the class. Check the recording for the demo . Check the Labware for the python code files

☐ Introduction

- ☐ Types of Robots
- ☐ Mobile Robots – Use case of Programmable solutions

☐ Components of Programmable Robotic System

- ☐ Perception
- ☐ Representation
- ☐ Motion

☐ Localization

- ☐ Taxonomy
- ☐ Types

☐ Mapping

- ☐ Feature Based
- ☐ Grid Based

☐ Path Planning

- ☐ Graph Based
- ☐ Sample Based
- ☐ Evolutional Techniques

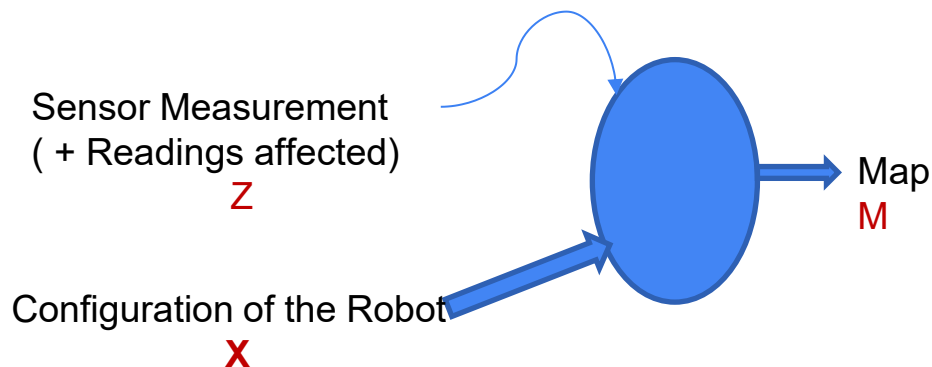
☐ Human – Robot Interaction

- ☐ Ethics
- ☐ Introduction to Visual Bot
- ☐ Introduction to Chat Bot

Mapping

Inverse Model for Mapping

If there is a sensor reads Z , what is the probability that there is a wall at D ?

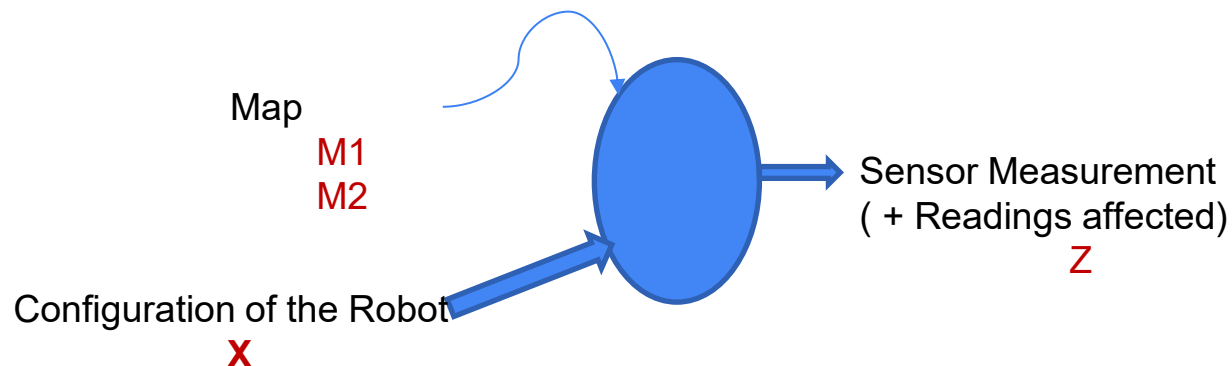


Given a measurement Z , how likely it would match to a sample map m ?
Analogy : Discriminative ML Models

$$P(M | Z, X)$$

Forward Model for Mapping

If there is a wall at distance D , what is the probability my sensor reads z ?



Given a hypothetical map m , how likely am I to see these measurements Z ?

Analogy : Generative ML Model

$$P(Z | M, X)$$

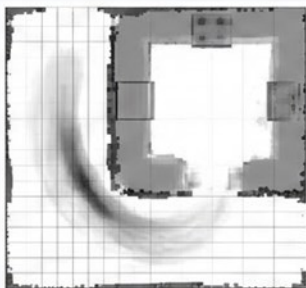
Occupancy Grid Mapping

Summary - Mapping Algorithms



Traditional Per-Grid Estimation (Bayesian)

- ✓ Updates each cell independently
- ✓ Log-odds recursive formula for recursive Bayes

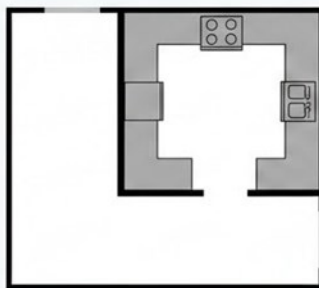


Traditional Hallway Mapping



Maximum A Posteriori (MAP) Optimization

- ⊕ Maximizes entire joint posterior $P(m|Z, X)$
- ⊕ Solves for globally consistent map m^*



MAP Optimized Grid



Hill-Climbing Optimization (with MAP)

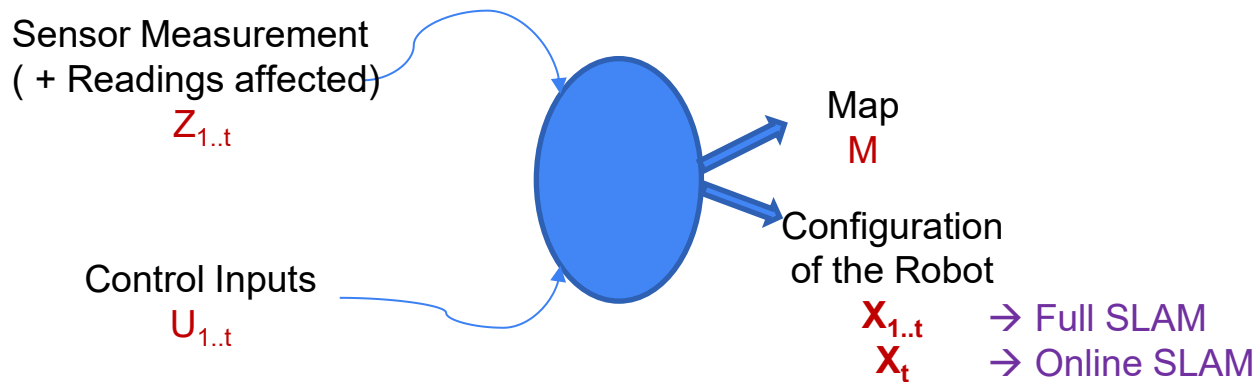
- ⊕ Iterative heuristic search on a guess map
- ⊕ Active subset search only on line-of-sight cells
- ⊕ Mobile robot 'flipping' only illuminated cells from free to occupied



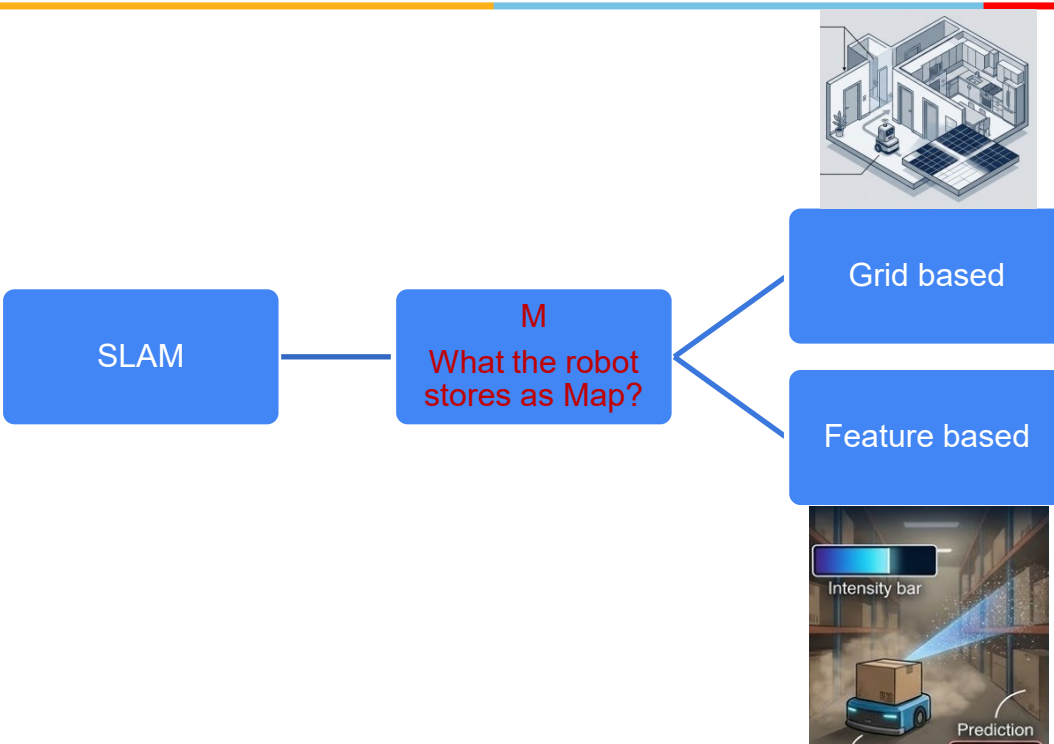
Hill-Climbing Local Update

Simultaneous Localization and Mapping

SLAM



SLAM



Occupancy grid SLAM or Dense map SLAM

- Each cell stores a probability: occupied/free/unknown
- The robot predicts its motion and aligns sensor data (e.g. LIDAR scans) with the grid
- **Hector-SLAM, Tiny-SLAM, Cartographer**

Landmark based SLAM or Sparse map SLAM

- Robot localizes itself relative to the landmarks
- Each measurement tries to identify (associate) which landmark it belongs to.
- **EKF-SLAM, EIF-SLAM, SEIF-SLAM, Fast-SLAM, GraphSLAM, Visual SLAM**

SLAM

SLAM

M
What the robot
stores as Map?

Grid based

Feature based

Histogram
eg., EKF-SLAM

X
How uncertainty
is represented?

Gaussian
eg., EKF-SLAM

Particle Filter
eg., Fast-SLAM

Note : Under the scope of our course coverage, only filtering based estimation is included. Interested students may explore graph based optimization for SLAM

- Is the state space naturally discrete or need a continuous Gaussian assumption?
- Is everything linear & Gaussian in a perfectly unambiguous environment?
- Is there a need for MHT in multimodal

